

CMPUT 654: Mathematical Foundations of Modern AI Systems

Fall 2026

Lecture 1: Decoder transformers and autoregressive inference

September 1, 2026

*Lecturer: Csaba Szepesvári**Scribe: instructor model note*

Purpose of this note. This is an example of the expected scribe-note style: record what happened in the lecture, define the mathematical objects precisely, and distinguish the main calculation from explanatory remarks. It is an exposition, not a transcript.

Delivery note. Administrative material occupied a substantial part of the meeting. The technical content was how a decoder transformer performs inference. Training was described verbally; no training objective, loss, or gradient equation was presented. The motivation for learning and the first learning-theoretic results were postponed to Lecture 2.

1.1 The inference problem

Given a token prefix $x_{<t} = (x_0, \dots, x_{t-1})$, a language model returns a categorical distribution for the next token:

$$x_{<t} \mapsto p_\theta(\cdot \mid x_{<t}). \quad (1.1)$$

An inference algorithm selects x_t from this distribution, appends it to the prefix, and repeats. Thus the model and the rule used to decode from it are distinct objects.

Let the vocabulary have size V , and identify token $x_t \in [V]$ with the standard basis vector $e_{x_t} \in \{0, 1\}^V$. Include a beginning-of-sequence token x_0 . For T prediction positions, define $Z \in \{0, 1\}^{T \times V}$ by

$$Z_{t,:} = e_{x_{t-1}}^\top, \quad t = 1, \dots, T. \quad (1.2)$$

With an embedding matrix $E \in \mathbb{R}^{V \times d}$ and additive absolute-position vectors $P \in \mathbb{R}^{T \times d}$, the input is

$$H^0 = ZE + P, \quad H_{t,:}^0 = E_{x_{t-1},:} + P_{t,:}. \quad (1.3)$$

The one-hot notation makes the map explicit; an implementation performs a row lookup in E .

Where is position added? Equation (1.3) describes additive absolute positions. They are added once, before the first transformer block; they are not added again in every block. Other position mechanisms enter the computation differently. RoPE applies a position-dependent rotation to queries and keys in each attention layer that uses RoPE. A relative-position bias is likewise added to the attention logits in each layer that uses it. These are alternative architectural choices; the symbol P in (1.3) does not also stand for RoPE.

1.2 One fully specified decoder block

Suppose $H^\ell \in \mathbb{R}^{T \times d}$ enters layer ℓ . Consider a pre-normalized block with h attention heads of width d_h . For head $a \in [h]$, let

$$U = \text{LN}_{\ell,1}(H^\ell), \quad (1.4)$$

$$Q_a = UW_{\ell,a}^Q, \quad K_a = UW_{\ell,a}^K, \quad V_a = UW_{\ell,a}^V, \quad (1.5)$$

$$A_a = \text{softmax}_{\text{row}}\left(\frac{Q_a K_a^\top}{\sqrt{d_h}} + M\right), \quad O_a = A_a V_a, \quad (1.6)$$

where

$$W_{\ell,a}^Q, W_{\ell,a}^K, W_{\ell,a}^V \in \mathbb{R}^{d \times d_h}, \quad M_{ij} = \begin{cases} 0, & j \leq i, \\ -\infty, & j > i. \end{cases} \quad (1.7)$$

The softmax acts row by row. The causal mask makes the attention weight of every future position exactly zero. Concatenate the head outputs and apply the first residual connection:

$$G = H^\ell + [O_1 \mid \cdots \mid O_h] W_\ell^O, \quad W_\ell^O \in \mathbb{R}^{hd_h \times d}. \quad (1.8)$$

The positionwise feed-forward sublayer and second residual connection give

$$R = \text{LN}_{\ell,2}(G), \quad (1.9)$$

$$H^{\ell+1} = G + \phi\left(RW_\ell^1 + \mathbf{1}(b_\ell^1)^\top\right) W_\ell^2 + \mathbf{1}(b_\ell^2)^\top, \quad (1.10)$$

with

$$W_\ell^1 \in \mathbb{R}^{d \times d_{\text{ff}}}, \quad W_\ell^2 \in \mathbb{R}^{d_{\text{ff}} \times d}, \quad b_\ell^1 \in \mathbb{R}^{d_{\text{ff}}}, \quad b_\ell^2 \in \mathbb{R}^d. \quad (1.11)$$

For example, $\phi(z) = z\Phi(z)$ coordinatewise is GELU. Practical models change normalization, gated MLPs, attention-head organization, and position handling. The equations specify one complete pedagogical block, not the unique modern design.

The key invariant is causal dependence: after L blocks, row $h_t^L = H_{t,:}^L$ depends only on $x_{<t}$.

1.3 From a hidden vector to the next-token distribution

Let $W_U \in \mathbb{R}^{d \times V}$ and $b_U \in \mathbb{R}^V$. The final row is converted to logits and then to probabilities:

$$z_t = h_t^L W_U + b_U, \quad (1.12)$$

$$p_\theta(x_t = v \mid x_{<t}) = \frac{\exp(z_{t,v})}{\sum_{u=1}^V \exp(z_{t,u})}. \quad (1.13)$$

Weight tying sets $W_U = E^\top$; it is common but not logically required.

The single-token path is therefore

$$e_{x_{t-1}} \longrightarrow E_{x_{t-1},:} \longrightarrow h_t^L \longrightarrow z_t \longrightarrow p_\theta(\cdot \mid x_{<t}). \quad (1.14)$$

The contextual vector h_t^L also depends on the earlier tokens through the masked attention blocks.

1.4 Autoregressive generation

A sampled decoder draws

$$X_t \sim p_\theta(\cdot \mid X_{<t}), \quad (1.15)$$

whereas greedy decoding chooses

$$X_t \in \arg \max_v p_\theta(v \mid X_{<t}). \quad (1.16)$$

In either case the selected token is appended and the procedure repeats until a stopping rule fires. Implementations cache keys and values from earlier positions, so each generation step need not recompute the entire prefix.

Greedy decoding is neither sampling from the model nor, in general, global optimization of the probability of the complete sequence. A locally most probable token can lead to poor later continuations.

1.5 What was said about training

Training was described only in words. The system is shown many text prefixes together with the tokens that actually followed them. Its predictions are compared with those next tokens, and the shared weights are modified to improve the predictions across the data. Lecture 3 will express this description using probabilistic models, log loss, maximum likelihood, KL divergence, and compression.

1.6 Questions carried into homework

1. **Relative position.** For block-rotation matrices $\mathcal{R}(t)$, prove

$$\mathcal{R}(s)^\top \mathcal{R}(t) = \mathcal{R}(t - s)$$

and hence the RoPE identity

$$(\mathcal{R}(s)q_s)^\top (\mathcal{R}(t)k_t) = q_s^\top \mathcal{R}(t - s)k_t.$$

2. **Greedy decoding.** Construct a family of conditional distributions for which repeated tokenwise argmax returns a sequence whose probability is exponentially smaller in its length than that of the maximum-probability sequence.

1.7 Take-home messages

- A decoder transformer maps a prefix to a categorical distribution by embedding tokens, applying causally masked attention and positionwise transformations, and using a softmax readout.
- Absolute position embeddings are added once at the input. RoPE and relative biases instead modify attention in every layer that uses them.
- The learned conditional distribution and the decoding algorithm are different objects. Tokenwise argmax does not solve a general sequence-level optimization problem.
- This lecture described inference. It did not yet explain why learning produces useful parameters.

References

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. Gomez, L. Kaiser, and I. Polosukhin. Attention Is All You Need. *NeurIPS*, 2017. <https://arxiv.org/abs/1706.03762>.
- [2] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu. RoFormer: Enhanced Transformer with Rotary Position Embedding. 2021. <https://arxiv.org/abs/2104.09864>.